Weekly Report – Week 2

Topic: Threat Detection & IOC Extraction

Objective

The objective of this week was to understand and identify Indicators of Compromise (IOCs) by analyzing system logs and phishing email data.

The focus was on detecting suspicious activities such as abnormal process execution, malicious command-line behavior, and phishing indicators.

Additionally, the goal was to gain hands-on experience in extracting and analyzing IOCs including file hashes, IP addresses, domains, and URLs.

Activities Performed

1. Sysmon Log Analysis

The file **`sysmon_logs.csv`** was analyzed using SIEM queries. The following steps were performed:

- Opened and reviewed system logs
- Searched for suspicious processes such as PowerShell and command prompt
- Identified unusual command-line arguments such as encoded commands and hidden execution
- Examined parent-child process relationships to detect anomalies

Special attention was given to processes like PowerShell and cmd.exe due to their frequent use in attacks.

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2. Detection of Suspicious Activity

Several suspicious behaviors were identified during analysis:

1. PowerShell executed with encoded commands (**`-enc`**).
2. Hidden execution flags (**`-nop`, `-w hidden`**).
3. Use of credential dumping tools such as Mimikatz.
4. Unusual parent processes such as winlogon.exe and svchost.exe launching command-line tools.

These behaviors indicate possible malicious activity or system compromise.

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3. IOC Extraction from Logs

From the identified suspicious events, the following IOCs were extracted:

- Process Name
- Command Line Arguments
- Parent Process
- File Hash (SHA256)

These indicators help in tracking and identifying malicious activity across systems.

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4. Phishing Email Analysis

The file **`fake_phishing_email.txt`** was analyzed to identify phishing indicators. The following elements were extracted:

- Sender email address
- Source IP address
- Domain name
- Malicious URL
- Attachment file name

The email contained characteristics of phishing such as urgency, fake security alerts, and malicious links.

IOC Extraction Task:

Step1: Identify the Suspicious Activity(Sysmon logs)

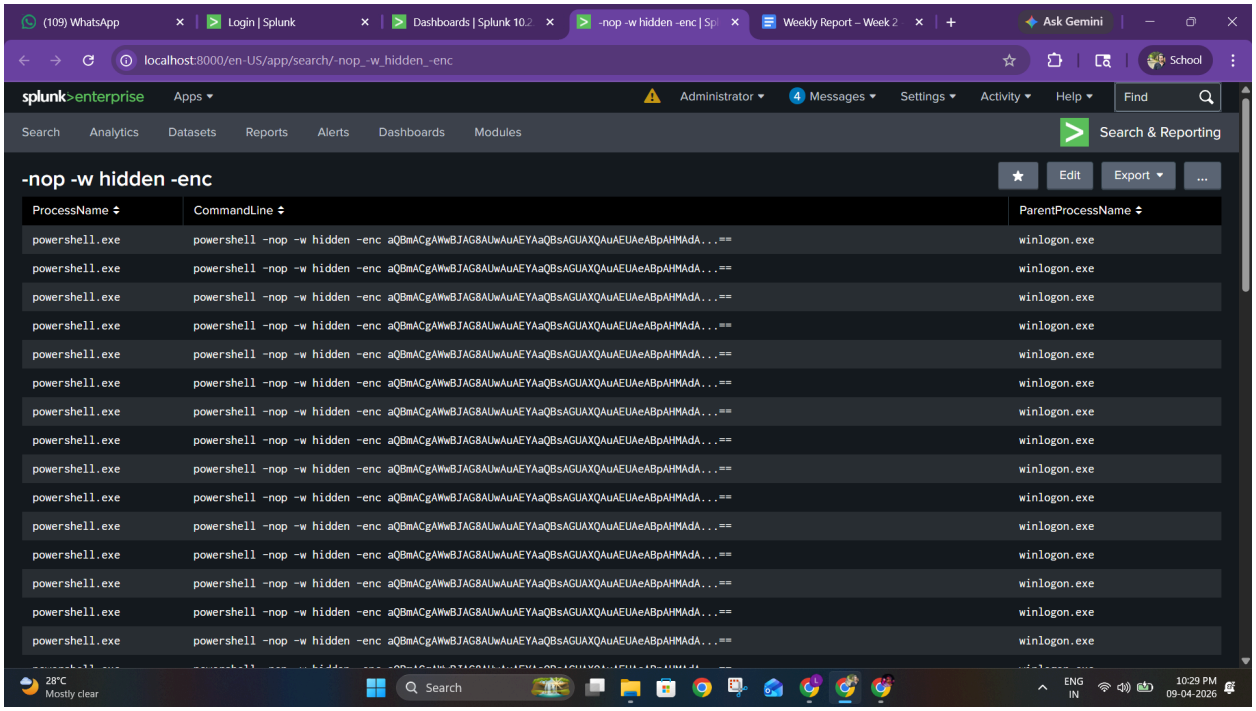
Suspicious Processes identified:

- powershell.exe (encoded and hidden execution)
- cmd.exe (used to execute malicious commands)
- mimikatz.exe (credential dumping tool)

Step 2: Extract IOC'S from Logs

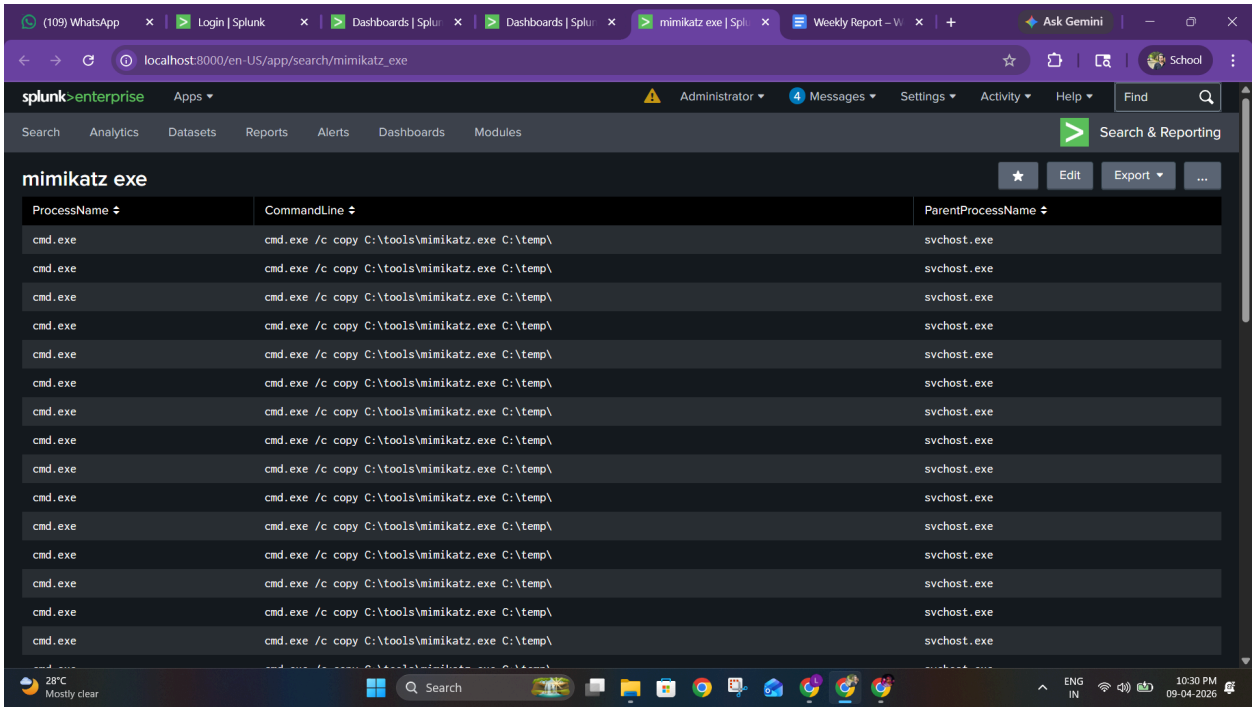
IOC 1

- Process: powershell.exe
- CommandLine: powershell -nop -w hidden -enc JAB...
- ParentProcess: winlogon.exe



IOC 2

- Process: cmd.exe
- CommandLine: cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\
- ParentProcess: svchost.exe



ProcessName	CommandLine	ParentProcessName
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe
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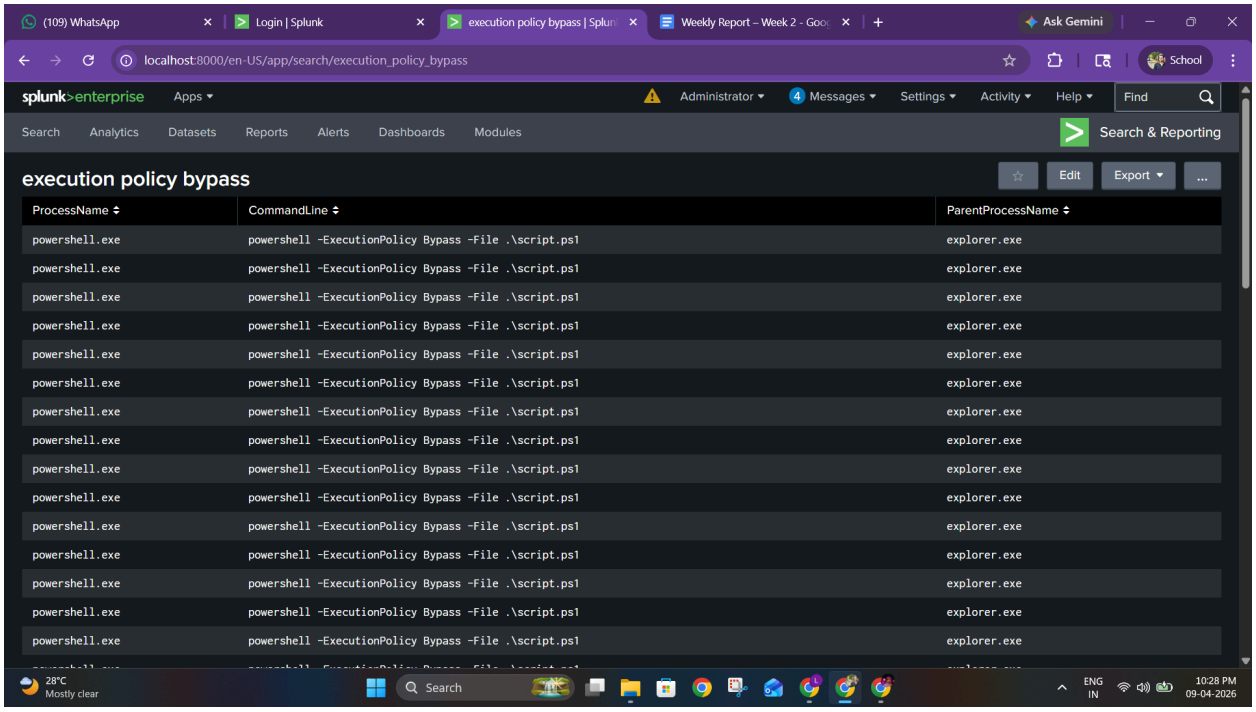
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe
cmd.exe	cmd.exe /c copy C:\tools\mimikatz.exe C:\temp\	svchost.exe

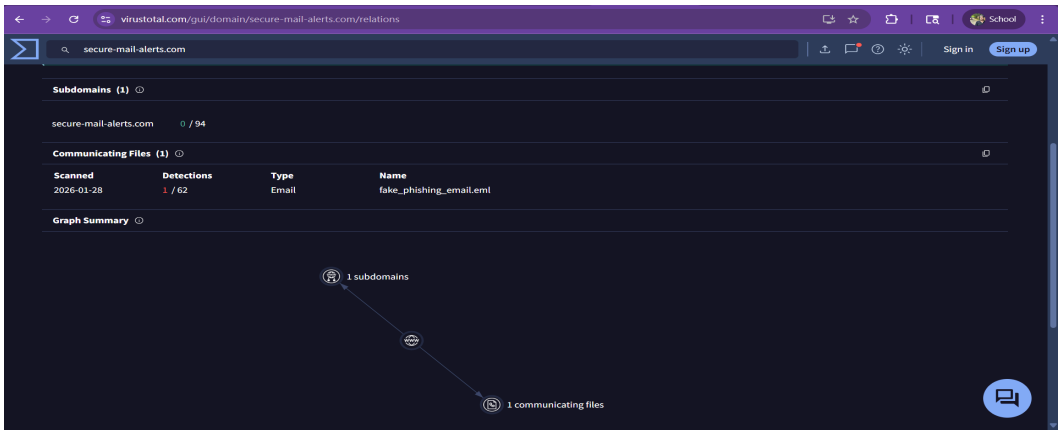
IOC 3

- Process: powershell.exe
- CommandLine: powershell -ExecutionPolicy Bypass -File script.ps1
- ParentProcess: explorer.exe



Step 3: Extract IOC'S from Phishing Email

- Sender Email: support@secure-mail-alerts.com
- IP Address: 185.234.217.101
- Domain: secure-mail-alerts.com
- URL: http://secure-mail-alerts.com/verify-account/login.php?session=abc123
- Attachment: invoice_July2025.exe



Step 4: Categorize IOCs

IOC	TYPE
SHA256 Hash	File IOC
185.234.217.101	IP IOC
secure-mail-alerts.com	Domain IOC
Phishing URL	URL IOC
invoice_July2025.exe	File IOC

verification: IOCs were checked using VirusTotal to confirm malicious indicators.

Key Findings

- Detection of **Mimikatz** suggests credential theft activity
- Suspicious domains and URLs were identified, indicating phishing attempts
- Malicious attachment (`.exe` file) indicates possible malware delivery mechanism
- Unusual parent-child process relationships highlight abnormal system behavior

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IOC Verification

The extracted IOCs such as IP address, domain, and file indicators were verified using **VirusTotal**.

The analysis confirmed that:

- The domain and IP are potentially malicious
- The attachment file is likely harmful
- The URL is associated with phishing activity

Learning Outcomes

During this week, the following concepts and skills were learned:

- Understanding of Indicators of Compromise (IOCs)
- Identification of suspicious process execution patterns
- Analysis of command-line behavior in system logs
- Detection of phishing emails using email headers and content
- Use of threat intelligence tools for IOC validation
- Importance of behavior-based detection over signature-based detection

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Recommendations

Based on the analysis, the following recommendations are suggested:

- Avoid opening email attachments from unknown sources
- Block malicious IP addresses and domains at network level
- Monitor PowerShell usage for encoded or hidden execution
- Implement endpoint detection and response (EDR) solutions
- Conduct regular log monitoring and threat hunting activities
- Educate users about phishing attacks and social engineering techniques

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Conclusion

The activities performed during Week 2 provided practical exposure to threat detection and IOC extraction techniques. The analysis of system logs and phishing emails helped in identifying multiple indicators of compromise, demonstrating how attackers operate and how such activities can be detected. This exercise improved understanding of real-world cybersecurity scenarios and strengthened analytical skills required for a SOC analyst role.